CSE 231: ANF Conversion

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Normal Forms

A key step in compilation is the conversion to **A-Normal Form** (**ANF**) where, informally speaking, each call or primitive operation's arguments are **immediate** values, i.e. constants or variable lookups whose values can be loaded with a single machine instruction.

```
(* (* (+ 2 3) (- 12 4)) (+ 7 8))
```

QUIZ: What is the A-Normal Form of the above?

```
(let* ((t1 (+ 2 3))
       (t2 (- 12 4))
       (t3 (+ 7 8))
       (t4 (* t1 t2)))
  (* t4 t3))
```

Source Language

Lets start with a source language that we wish to work with:

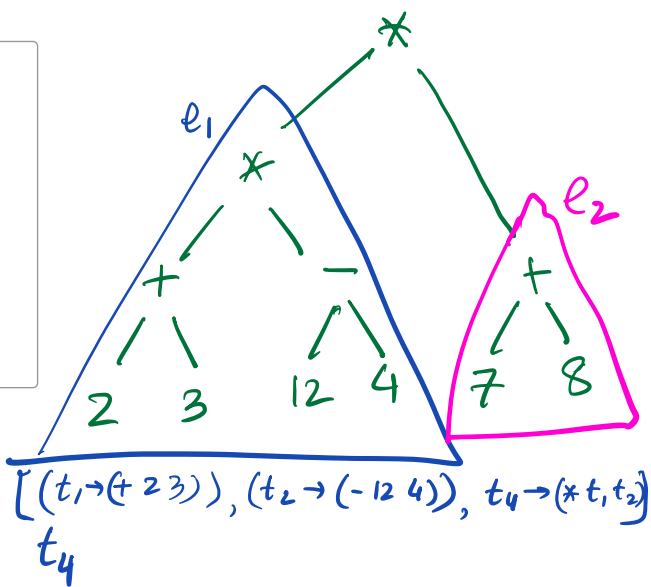
```
type Id = String;

enum Op { Add, Sub, Mul }

enum Exp {
  Var(String),                    // vars: x, y, ...
  Num(i32),                       // numbers: 0, 1,...
  Bin(Op, Box<Exp>, Box<Exp>), // binops
  Let(Id, Box<Exp>, Box<Exp>), // let-binders
}
```

For example, the source expression above corresponds to:

```
// (* (* (+ 2 3) (- 12 4)) (+ 7 8))
fn exp0() -> Exp {
  bin(Mul, bin(
             Mul,
             bin(Add, num(2), num(3)),
             bin(Sub, num(12), num(4)),
           ),
           bin(Op::Add, num(7), num(8)),
    )
}
```



A-Normal Form

Before we can describe a *conversion* to A-Normal Form (ANF), we must pin down what ANF *is*. Our informal description was:

Immediate Values: "*constants or variable lookups whose values can be loaded with a single machine instruction*"

ANF Expressions: "*each call or primitive operation's arguments are **immediate** values, i.e. constants or variable lookups*"

We can encode these as Rust predicates `is_imm` and `is_anf`

```
impl Exp {
  fn is_imm(&self) -> bool {
    match self {
    - Var(_) => true,
    . Num(_) => true,
      Bin(_, _, _) => false,
      Let(_, _, _) => false,
    }
  }

  fn is_anf(&self) -> bool {
    match self {
      Var(_) => true,
      Num(_) => true,
      Bin(_, e1, e2) => e1.is_imm() && e2.is_imm(),
      Let(_, e1, e2) => e1.is_anf() && e2.is_anf(),
    }
  }
}
```

We *could* define brand new types, say `ImmExp` and `AnfExp`, whose values correspond to *immediate* and *ANF* terms. Unfortunately, doing so leads to a bunch of code duplication, e.g. duplicate printers for `Exp` and `AnfExp`. Try it, as an exercise.

ANF Conversion: Intuition

Recall that our goal is to convert expressions like

```
(* (* (+ 2 3) (- 12 4)) (+ 7 8))
```

into a let-bound expression of the form

```
(let (?tmp0 (+ 2 3))
  (let ?tmp1 = (- 12 4) in
    (let ?tmp2 = (* ?tmp0 ?tmp1) in
      (let ?tmp3 = (+ 7 8) in
        (* ?tmp2 ?tmp3)))))
```

Generalizing a bit, we want to convert

```
(+ e1 e2)
```

into a sequence of let-bindings that introduce the names needed to make the arguments `e1` and `e2` immediate

```
(let (x1 a1) ... (let (xn an)
   (let (x1' a1') ... (let (xm' am')
     (+ v1 v2) ...) ... )
```

where `v1` and `v2` are immediate, and each `ai` is ANF.

Forcing Arguments to be Immediate

The key requirement is a way to **force** arbitrary *argument expressions* like `e1` into **a pair**:

- a **vector of bindings** `[(x1, a1), ..., (xn, an)]` where each `ai` is `Anf`, and
- an **immediate expression** `v1` of type `Imm`.

so `e1` is *equivalent* to `(let (x1 a1) ... (let (xn an) v1))`.

Thus, we need a method to **make arguments immediate**, yielding a top-level conversion method:

```
fn to_imm(&self,
    count: &mut usize,
    binds: &mut Vec<(Id, Anf)>
) -> Imm;

fn to_anf(&self, count: &mut usize) -> Anf;
```

QUIZ: Making Arguments Immediate

```
fn to_imm(&self, count: &mut usize,
  binds: &mut Vec<(Id, Anf)>) -> Imm {
  match self {
    Exp::Var(x) | Exp::Num(n) => self.clone()
    Exp::Bin(op, e1, e2) => {
        let i1 = e1.to_imm(count, binds);
        let i2 = e2.to_imm(count, binds);
        let id = count.fresh();
        binds.push((id, Bin(op, i1, i2)))
    } Valid)
    Exp::Let(..) => {
        let a = self.to_anf(count);
        let id = count.fresh();
        binds.push((id, a));
        Var(id)
    }
  }
}
```

- **Numbers and variables** are already immediate, and are returned directly.

- **Binary operators** recursively make their operands immediate, generate a fresh binder for the operation, and return the fresh variable.
- **Let-binders** are converted to ANF and assigned to a fresh binder.

Fresh Variables

We use a `count: &mut usize` to produce `fresh` names:

```
fn fresh(count: &mut usize) -> Id {
  let name = format!("?tmp{}", count);
  *count += 1;
  name.to_string()
}
```

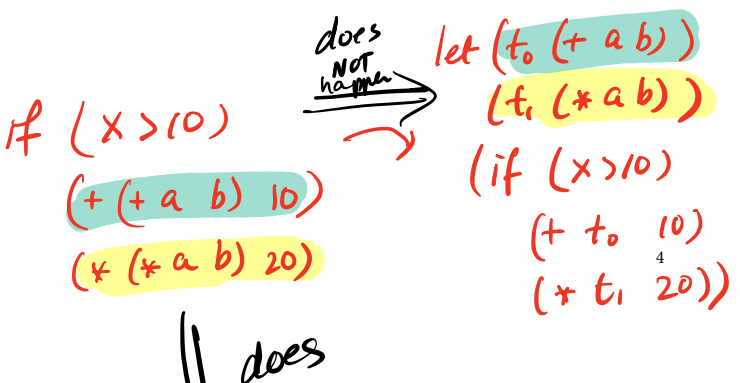
QUIZ: Top-Level Conversion: `to_anf`

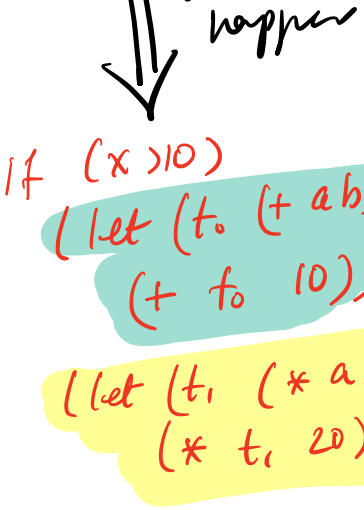
```
fn to_anf(&self, count: &mut usize) -> Anf {
  match self {
    Exp::Var(x) | Exp::Num(n) => self.clone()

    Exp::Let(x, e1, e2) => {
    }
    Exp::Bin(op, e1, e2) => {
    }
  }
}
```

In `to_anf`, the real work happens inside `to_imm` which takes an arbitrary *argument* expression and makes it **immediate** by generating temporary (ANF) bindings.

The resulting bindings (and immediate values) are composed by popping from the `binds` vector and wrapping the result with `let_` to stitch them into a single `Anf` expression.






```
(let (tmp1 (+ 30))
  (* tmp0 tmp1)))
```

```
if (x > 10)
(let (t0 (+ a b))
  (+ t0 10))
(let (t1 (* a b))
  (* t1 20))
```